

SOUTH ALEUTIANS BUOY 46003, USAWMO: **992170**Lat: **51.831N**Lon: **155.850W**Elev: **10**StdP: **14.69**Time Zone: **-10.00 (W10)**Period: **86-99**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.1	29.7	N/A	N/A	N/A	N/A	N/A	N/A	38.1	35.5	36.7	35.9	19.4	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.9	55.7	N/A	55.1	N/A	54.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	17.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.5	34.1	31.7	DB	24.6	57.4	2.0	3.1	23.2	59.6	22.0	61.4	20.9	63.1	19.4	65.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	37.7	36.9	37.1	37.8	41.1	44.7	49.2	52.4	51.5	46.6	41.8	39.3	
	DBStd	6.01	2.93	3.67	2.48	1.94	1.48	1.80	2.04	1.45	2.11	2.67	3.36	2.82	
	HDD50	2682	382	366	399	366	275	159	41	0	10	107	245	332	
	HDD65	8011	847	786	864	816	740	609	491	392	405	569	695	797	
	CDD50	148	0	0	0	0	0	0	15	74	56	3	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	17.3	18.7	19.8	17.6	16.9	14.6	14.1	13.2	14.9	18.0	19.1	20.6	20.9	
Precipitation	PrecAvg	45.4	4.4	3.8	3.1	2.4	2.5	2.8	3.5	4.1	4.9	4.8	4.3	5.0	
	PrecMax	55.4	6.7	5.9	4.4	4.3	4.0	5.3	5.3	7.1	8.7	7.6	6.9	8.5	
	PrecMin	35.3	1.9	0.7	1.4	0.6	0.9	1.1	2.0	1.3	3.0	2.7	2.2	2.8	
	PrecStd	4.4	1.1	1.3	0.9	0.8	0.7	0.9	0.9	1.4	1.2	1.3	1.1	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	43.0	42.1	42.1	46.1	50.6	54.9	57.3	56.5	54.3	50.2	45.4	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	42.4	42.2	41.4	41.4	44.5	49.4	53.7	56.2	55.5	52.2	49.1	44.3	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	41.8	41.5	41.0	41.0	43.8	48.3	52.8	55.3	55.1	51.0	48.0	43.5	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	41.3	41.2	40.5	40.4	43.2	47.3	52.1	54.7	54.5	50.2	46.6	42.8	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.6	3.3	2.9	2.1	1.9	1.9	1.9	2.4	3.0	3.8	3.8	
		MCDBR	3.4	3.7	3.6	3.0	2.3	2.2	2.0	2.2	2.7	3.3	3.3	4.2	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.334	0.352	0.378	0.428	0.408	0.394	0.400	0.398	0.394	0.368	0.347	0.338		
	taud	2.285	2.299	2.229	2.109	2.244	2.367	2.409	2.415	2.329	2.291	2.220	2.203		
	Ebn at Noon	207	239	255	255	268	273	269	263	248	228	195	178		
	Edh at Noon	21	28	36	45	41	37	35	33	32	27	22	20		
All-Sky Solar Radiation	RadAvg	235	463	827	1176	1390	1355	1260	1082	894	575	294	183		
	RadStd	21	35	58	70	129	110	86	102	55	41	26	18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page